

The Neural Correlates of “Food Addiction”

Context Research has implicated an addictive process in the development and maintenance of obesity. Although parallels in neural functioning between obesity and substance dependence have been found, no studies have examined the neural correlates of addictive-like eating behavior. Objective To test the hypothesis that elevated “food addiction” scores are associated with similar patterns of neural activation as substance dependence. Design Between-Subjects fMRI study. Participants Forty-eight healthy adolescent females ranging from lean to obese recruited for a healthy weight maintenance trial. Main Outcome Measure The relation between elevated “food addiction” scores and blood oxygen level-dependent fMRI activation in response to receipt and anticipated receipt of palatable food (chocolate milkshake). Results Food addiction scores ($N = 39$) correlated with greater activation in the anterior cingulate cortex (ACC), medial orbitofrontal cortex (OFC), and amygdala in response to anticipated receipt of food ($P < 0.05$, false-discovery rate (FDR) corrected for multiple comparisons in small volumes). Participants with higher ($n=15$) versus lower ($n=11$) food addiction scores showed greater activation in the dorsolateral prefrontal cortex (DLPFC) and the caudate in response to anticipated receipt of food, but less activation in the lateral OFC in response to receipt of food ($pFDR < 0.05$). Conclusions Similar patterns of neural activation are implicated in addictive-like eating behavior and substance dependence; elevated activation in reward circuitry in response to food cues and reduced activation of inhibitory regions in response to food intake.